



Reproduction of antibody levels following five distinct ChAdOx1 nCoV-19 vaccine regimens through virtual patient cohorts

Maicom Peters Xavier^{*}, Rodrigo Weber dos Santos, Marcelo Lobosco^{ID}

Graduate Program on Computational Modelling, Rua José Lourenço Kelmer, s/n, Plataforma da Faculdade de Engenharia, Juiz de Fora, 36.036-330, Minas Gerais, Brazil

ARTICLE INFO

Keywords:

Computational vaccinology
Computational immunology
Computational modelling
COVID-19 vaccine

ABSTRACT

We present a mathematical model of the immune response elicited by the ChAdOx1 nCoV-19 vaccine across primary and booster immunisations. The model was calibrated with clinical cohort data. To assess population variability, we generated a Virtual Cohort (VC) of > 16,000 virtual patients by sampling biologically plausible parameter sets that reproduced observed antibody trajectories. In silico trials compared five adult vaccination scenarios (ages 18–55): a two-dose primary regimen, booster schedules at 8–12, 15–25, and 44–45 weeks, and an additional third-dose option. Model predictions were validated against longitudinal clinical data up to one year after booster administration. Analysis of the VC identified timing strategies for two- and three-dose regimens that promote sustained antibody responses. These results illustrate how computational models and virtual populations can inform vaccine scheduling and support optimisation of vaccination strategies.

1. Main

COVID-19, caused by SARS-CoV-2, had a profound global impact. The disease produced substantial mortality and severe complications, including pneumonia, acute respiratory distress syndrome (ARDS) and organ failure. In January 2020 the World Health Organization (WHO) declared COVID-19 a global health emergency because of its rapid international spread [1,2]. Remarkably, within 17 months the WHO authorised six vaccines with reported efficacies between 50% and 95% against symptomatic infection [3]. These vaccines include mRNA platforms (Pfizer–BioNTech, Moderna), inactivated-virus formulations (CoronaVac) and viral-vector vaccines (Oxford–AstraZeneca). Collectively they reduced severe disease, hospitalisations and mortality. Although the WHO reclassified COVID-19 in May 2023 as no longer a global health emergency, the disease remains a public-health concern.

Despite rapid progress in vaccine development, key questions remain. The durability of vaccine-induced immunity is uncertain, especially in the face of emerging variants. Optimal dosing intervals are still debated. Long-term adverse events and vaccine effectiveness in subpopulations (for example, immunocompromised individuals and children) require further clarification. Addressing these gaps is essential to refine vaccination strategies and to inform broader infectious-disease control.

Mathematical and computational models have played a central role in vaccine research [4–20]. These methods are widely used in virtual screening and reverse vaccinology and have been applied in clinical

development to predict dosing, administration frequency and immunity duration [21,22]. Mathematical frameworks also support target identification and guide drug and vaccine design.

A range of modelling approaches has been employed. Ordinary differential equations (ODEs) describe temporal immune dynamics and intercellular interactions [23–28]. Partial differential equations (PDEs) capture spatial and temporal aspects of immune responses [29–33]. Stochastic methods, cellular automata and agent-based models provide complementary insights into immune-system behaviour [34–40]. More recently, digital-twin approaches, borrowed from industry, have been adapted to biology to produce virtual replicas of physiological systems. These digital twins can simulate responses under varied conditions and hold promise for personalised vaccination strategies [41–44].

Modelling of SARS-CoV-2 vaccine dynamics is still emerging. One recent study fitted one- and two-compartment models and a double-exponential model to antibody decay after BNT162b2 (Pfizer–BioNTech) vaccination [45]. That work, however, omitted long-term immune mechanisms such as memory B cells and validated results only over a 40-week window. Such a horizon is insufficient to characterise longer-term antibody persistence. The study further predicted antibody decline to near zero around 100 weeks, but did not test that projection across different vaccination regimens.

Another study used an immunostimulation/immunodynamic (IS/ID) approach [46] with sigmoidal and exponential functions to identify

^{*} Corresponding author.

E-mail addresses: maicom.peters@estudante.ufjf.br (M.P. Xavier), rodrigo.weber@ufjf.br (R.W. dos Santos), marcelo.lobosco@ufjf.br (M. Lobosco).

minimal effective vaccine doses [47]. The analysis covered four vaccines (ChAdOx1 nCoV-19, BNT162b2, Ad26.COV2.S and CoronaVac) in adults and elderly cohorts and suggested dose-specific adjustments. Yet the model's limited mechanistic scope reduced its ability to capture the full complexity of the immune response and thus constrained predictive power.

Here we present a mechanistic ODE model that extends our prior computational framework [22,48] to describe immune responses to the ChAdOx1 nCoV-19 vaccine. Our original framework was developed for live-attenuated vaccines [22] and has now been adapted to reflect viral-vector immunology. We recalibrated the model using antibody data from individuals vaccinated under five distinct schedules [49]. The model reproduces cohort-measured antibody levels at all evaluated time points. An independent study later confirmed sustained antibodies one year after booster in the same cohort [50], a result consistent with our projections.

To quantify population variability, we built a Virtual Cohort (VC) anchored to empirical IgG data from a reference calibration set of 115 individuals [49]. In this paper, a *virtual patient cohort* refers to an ensemble of parameter vectors $\theta_{j=1}^N$ that (i) produce model trajectories consistent with the clinical calibration data according to the acceptance criteria described in the next sections, and (ii) satisfy pre-specified physiological plausibility constraints where applicable. Each virtual patient is therefore a mechanistic instantiation of plausible within-host kinetics (antigen, APCs, T-helper cells, B-cell compartments, plasma cells and IgG) under a given dosing schedule. Cohort-level summaries (e.g. median, interquartile range, and percentile bands) are computed across the simulated trajectories of these accepted parameter vectors and reported throughout the manuscript to represent population-level heterogeneity. The VC contains over 16,000 virtual patients generated by sampling parameter sets that closely match observed immune trajectories. By anchoring the VC to validated data and applying strict acceptance criteria, our virtual population remains grounded in real-world variability. This population supports robust sensitivity analyses and *in silico* trials across diverse vaccination scenarios, evaluated over a 500-day horizon.

Although this work employs a mechanistic ordinary-differential-equation framework with uncertainty quantification and optimisation, there exists a rich body of computational methods in allied areas that are relevant to vaccine and drug discovery. Examples include graph-based contrastive learning for drug–target affinity prediction (GraphCL-DTA) [51], positive–unlabelled learning frameworks (PUON) [52], self-supervised methods to mitigate label sparsity in drug–disease association tasks [53], neural metric and matrix factorisation techniques for drug repositioning (NMFDR, ANMF) [54,55], and multi-omics deconvolution approaches (OmicsTweezer) [56]. These techniques complement mechanistic modelling: they can provide statistical priors, informative latent features or improved parameter priors for Bayesian calibration, or they can help stratify patients for fitting and validation.

This paper is organised as follows. Section 2 describes the mathematical model, calibration procedures and computational implementation, including sensitivity and uncertainty analyses. Section 3 presents the computational results. Section 4 interprets the findings.

2. Methods

This section describes the methods used to reproduce antibody levels following distinct ChAdOx1 nCoV-19 vaccine regimens. Fig. 1 schematically summarises the adopted approach. First, cohort data were extracted from the literature [49,50] and adapted for use in simulations (Section 2.1). Next, we developed a new mathematical model (Section 2.2), based on a previously published model [22], to reproduce *in silico* the immune response elicited by the ChAdOx1 nCoV-19 vaccine.

We calibrated the model to one representative vaccine regimen reported in the literature [49,50]. Calibration used Differential Evolution [57], an optimisation algorithm that minimises the discrepancy

between model outputs and cohort data (see Section 2.3). Parameters were varied within predefined bounds. We retained all parameter sets whose model outputs produced errors below a predefined acceptance threshold; these retained sets form the Virtual Cohort (VC; see Section 2.3). The single best-fitting parameter set (the one that minimises the error) was saved as the reference fit. Parameters selected for the VC were validated using vaccine regimens different from the calibration regimen.

We performed a sensitivity analysis to identify the most influential parameters (Section 2.4). Uncertainty was propagated by perturbing the key parameters around the best-fit values to assess output variability. Finally, we ran an optimisation loop to identify candidate booster schedules (Section 2.6). The objective of this loop was to maximise the fraction of the virtual population that maintains antibody levels above a predefined protective threshold.

2.1. Cohort data and their adaptation for use in simulations

This study used data from the COV001 and COV002 trials conducted in the UK [49,50]. We focused on participants aged 18–55 who received one or two doses of ChAdOx1 nCoV-19 in phase 1/2. Some participants who received two doses (8–12 weeks apart) were followed for one year after the second dose, confirming antibody persistence [50]. Participants who initially received a single dose were given a delayed second dose at about 44–45 weeks after the first dose [49]. Those who had two doses received a third dose 28–38 weeks after the second dose (median 30 weeks) [49]. We compare these cohort measurements with numerical results generated by our model.

Antibody values are reported at several time points corresponding to prime–boost intervals of 8–12, 15–25 and 44–45 weeks [49]. Immunogenicity data were available for 261 individuals. Of these, 115 (44%) had a second dose at 8–12 weeks, 116 (44%) at 15–25 weeks, and 30 (11%) at 44–45 weeks [49]. Eighty participants who had previously received two doses subsequently received a third dose 28–38 weeks after the second dose [49]. The persistence study [50] does not report the exact number of participants (from the overall 704 enrolled) who received the standard 8–12-week interval.

The cohort reported T cell responses measured as spot-forming units (SFUs) per million peripheral blood mononuclear cells (PBMCs), but did not distinguish T cell subtypes (CD4⁺, CD8⁺, regulatory T cells). Therefore, although our model includes T helper cells, direct comparison with subtype-resolved T cell data is not possible.

The standard ChAdOx1 nCoV-19 dose contains 5×10^{10} viral particles. The model requires an initial vaccine concentration (particles per ml) after inoculation. We estimate this concentration by assuming homogeneous dilution in total body fluid. Total body water varies with age, sex and body composition, typically ranging from 50% to 70% of body mass [58,59]. We used Watson's formula [60] and assumed an average adult of height 1.70 m and mass 70 kg. The resulting average total body water fraction is approximately 0.53. Thus the initial adenovirus concentration is

$$V_0 = \frac{5 \times 10^{10}}{70 \times 1000 \times 0.53} \approx 1.35 \times 10^6 \text{ particles ml}^{-1}. \quad (1)$$

The source study [49] reports medians, geometric mean titres (GMT), and ranges, but does not provide measures of dispersion such as standard deviation or standard error. When dispersion was required for calibration or uncertainty propagation, we estimated the standard deviation from the reported range. As a transparent and reproducible default, we applied the Ramirez and Cox estimator [61], which adjusts the range-based estimate according to the sample size n :

$$\sigma \approx \frac{\text{sample range}}{3\sqrt{\ln n - 1.5}}, \quad (2)$$

where n denotes the sample size. Ramirez and Cox demonstrated that this estimator performs reliably for approximately normal data with

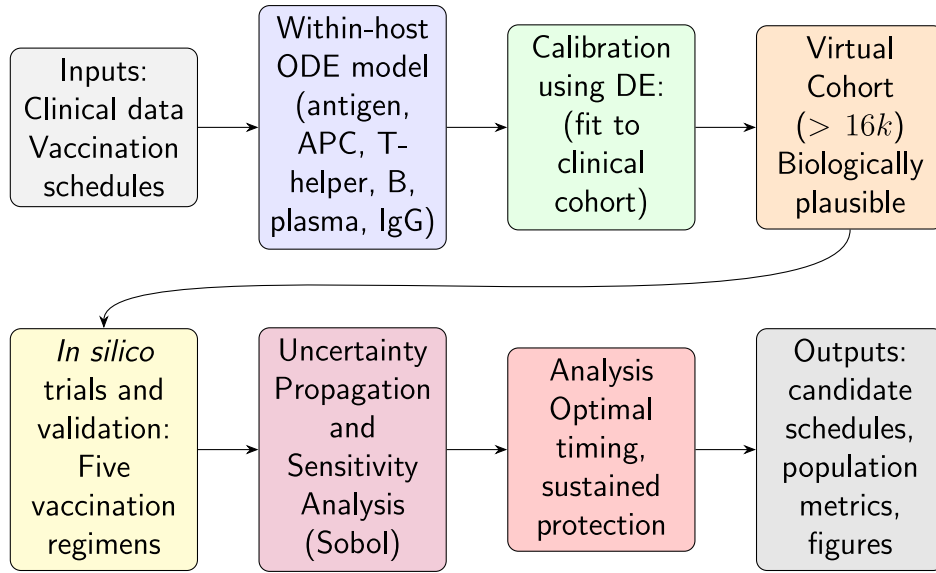


Fig. 1. Workflow of the within-host immunological model, showing the integration of clinical data, calibration, virtual cohort generation, uncertainty propagation, sensitivity analysis, and optimisation to identify effective vaccination schedules.

moderate sample sizes, and they also provide alternatives for small or strongly skewed samples. Although a fully Bayesian approach would represent a more principled treatment, it lies beyond the scope of this study. Given the absence of reported dispersion and the moderate sample sizes available, the range-based estimator provides a suitable and reproducible solution for the present work.

2.2. Mathematical model

To capture the immune response to vaccination, we developed a mechanistic model formulated as a system of eleven ordinary differential equations (ODEs). The model describes how different immunological components interact over time after administration of the ChAdOx1 nCoV-19 vaccine. Specifically, the model tracks the dynamics of: the vaccine virus (V); the viral spike glycoprotein (S); antigen-presenting cells (APCs), in their naïve (A_p) and mature (A_{pm}) forms; CD4⁺ T helper cells, in naïve (T_{hn}) and activated (T_{he}) states; B cells, subdivided into naïve (B_n), activated (B_a), and memory (B_m) compartments; plasma cells (P_c); and circulating IgG antibodies (IgG). These components collectively represent the cascade of immunological events that starts with antigen presentation and culminates in antibody production.

Biologically, after intramuscular administration, the adenoviral vector infects host cells, leading to expression of the spike glycoprotein. This antigen is then recognised by APCs, which process and present it to CD4⁺ T cells. Activated CD4⁺ T cells provide help for B cell activation and differentiation. B cells then generate plasma cells, which are responsible for secreting IgG antibodies, and memory B cells, which enable a faster and more robust response upon re-exposure to the antigen. Our model formalises this biological sequence using ODEs, where each term corresponds to a specific biological process: production, activation, proliferation, or natural decay.

The ChAdOx1 nCoV-19 vaccine employs a replication-deficient chimpanzee adenoviral vector engineered to express the spike protein [62]. Because the viral vector cannot replicate, viral load decreases over time without exponential growth. This is reflected in the virus equation (Eq. (3)):

$$\frac{d}{dt} V = -\delta_v V - \delta_{v_{igg}} V IgG. \quad (3)$$

The first term, $\delta_v V$, describes the natural clearance of viral particles and their elimination by innate immune mechanisms. The second term, $\delta_{v_{igg}} V IgG$, represents neutralisation of viral particles by IgG antibodies,

emphasising the role of humoral immunity in viral clearance. Compared to our previous work [48], this version omits an explicit innate clearance term and instead incorporates innate and natural decay into δ_v .

The viral spike protein (S) plays a central role, as it is the antigen that drives adaptive immune activation. Even though the vector does not replicate, infected cells continuously release spike protein, modelled in Eq. (4):

$$\frac{d}{dt} S = \alpha_s V - \delta_{s_{igg}} S IgG - \delta_{s_{ap}} S A_p, \quad (4)$$

The production term $\alpha_s V$ reflects spike protein synthesis proportional to viral presence. Neutralisation by IgG antibodies is represented by $\delta_{s_{igg}} S IgG$. Finally, uptake by naïve APCs, $\delta_{s_{ap}} S A_p$, initiates antigen presentation. In this context, A_p may represent dendritic cells or macrophages that internalise the antigen and subsequently mature.

APCs are central to bridging innate and adaptive immunity. Immature APCs (A_p) are maintained at a baseline level through homeostasis and are activated upon encountering spike protein (Eq. (5)):

$$\frac{d}{dt} A_p = \alpha_{ap}(A_{p0} - A_p) - \gamma_{s_{ap}} A_p S. \quad (5)$$

The first term restores APC levels towards a baseline A_{p0} , while the second term describes their activation upon antigen uptake. The resulting mature APCs (A_{pm}) are modelled in Eq. (6):

$$\frac{d}{dt} A_{pm} = \gamma_{s_{ap}} A_p S - \delta_{apm} A_{pm}. \quad (6)$$

Here, $\gamma_{s_{ap}} A_p S$ corresponds to antigen-induced maturation, and $\delta_{apm} A_{pm}$ accounts for their natural turnover. Mature APCs present antigen via MHC class II molecules and interact with naïve CD4⁺ T cells.

Naïve CD4⁺ T cells (T_{hn}) are maintained by homeostasis and become activated upon interaction with mature APCs (Eq. (7)):

$$\frac{d}{dt} T_{hn} = \alpha_{th}(T_{hn0} - T_{hn}) - \beta_{th} A_{pm} T_{hn}. \quad (7)$$

The activation term, $\beta_{th} A_{pm} T_{hn}$, represents APC-driven priming of naïve T cells into effector T cells (T_{he}). Their subsequent proliferation and decay are represented in Eq. (8):

$$\frac{d}{dt} T_{he} = \beta_{th} A_{pm} T_{hn} + \pi_{th} A_{pm} T_{he} - \delta_{the} T_{he}. \quad (8)$$

Thus, effector T cells increase both through activation of naïve cells and through their own proliferation (π_{th}), while their natural death

is captured by $\delta_{the} T_{he}$. Effector CD4⁺ T cells provide help for B cell differentiation.

Naïve B cells (B_n) are the source of both active B cells (B_a) and memory B cells (B_m). Their dynamics (Eq. (9)) include homeostatic maintenance, activation by T helper cells, and natural decay:

$$\frac{d}{dt} B_n = \alpha_{bn}(B_{n0} - B_n) - \beta_{bnba} B_n T_{he} - \beta_{bm} T_{he} B_n - \delta_{bn} B_n. \quad (9)$$

Differentiation into active (B_a) or memory (B_m) subsets depends on interactions with effector T cells. Active B cells, in turn, can differentiate into plasma cells or be reactivated during reinfection, as captured by Eq. (10):

$$\frac{d}{dt} B_a = \beta_{bnba} B_n T_{he} + \gamma_{bm} B_m S - \beta_{ba_{pc}} A_{pm} B_a - \delta_{ba} B_a. \quad (10)$$

Here, $\gamma_{bm} B_m S$ represents reactivation of memory B cells, while $\beta_{ba_{pc}} A_{pm} B_a$ denotes APC-assisted differentiation into plasma cells.

Memory B cells (B_m) ensure long-term protection and can both self-renew and differentiate. Their dynamics are shown in Eq. (11):

$$\begin{aligned} \frac{d}{dt} B_m &= \beta_{bm} T_{he} B_n + \frac{B_m}{k_0 + k_1 B_m} \left(1 - \frac{B_m}{B_{mmax}}\right) \\ &- \pi_{bm_{pc}} B_m S - \gamma_{bm} B_m S - \rho_{bm_{pc}} B_m - \delta_{bm} B_m. \end{aligned} \quad (11)$$

The term

$$\frac{B_m}{k_0 + k_1 B_m} \left(1 - \frac{B_m}{B_{mmax}}\right)$$

encodes a saturating, self-limited expansion of memory B cells. When the memory pool is small relative to the characteristic scale k_0/k_1 , the fraction $\frac{B_m}{k_0 + k_1 B_m}$ is approximately $\frac{B_m}{k_0}$, and the term behaves like a conventional density-dependent (near-exponential) growth term with effective rate $1/k_0$. As the memory pool increases, however, the fraction saturates towards the value $1/k_1$, so that further increases in B_m do not proportionally elevate the instantaneous production rate; this models a throughput limit imposed by physiological constraints (for example, limited survival niches, cytokine availability, or metabolic resources). The multiplicative factor $(1 - B_m/B_{mmax})$ introduces an explicit carrying capacity, ensuring that net growth vanishes when the memory pool reaches B_{mmax} . Together, these factors produce biologically plausible behaviour: rapid expansion when memory B cells are scarce, tapering growth as resource constraints emerge, and an eventual stabilisation of the population at a physiologically realistic plateau. Other terms represent differentiation into plasma cells (fast response via $\pi_{bm_{pc}} B_m S$ or steady replacement via $\rho_{bm_{pc}} B_m$), reactivation, and decay.

Plasma cells (P_c), the antibody-producing factories, are modelled in Eq. (12):

$$\frac{d}{dt} P_c = \beta_{ba_{pc}} A_{pm} B_a + \pi_{bm_{pc}} B_m S + \rho_{bm_{pc}} B_m - \delta_{pc} P_c. \quad (12)$$

They arise from both active and memory B cells, and their numbers decline naturally at rate δ_{pc} .

Finally, IgG antibodies (IgG) are secreted by plasma cells, providing systemic protection (Eq. (13)):

$$\frac{d}{dt} IgG = \pi_{pc} P_c - \delta_{IgG} IgG. \quad (13)$$

The production rate $\pi_{pc} P_c$ links plasma cell numbers to IgG secretion, while $\delta_{IgG} IgG$ represents antibody clearance. Since the available cohort data report IgG titres only [49], our model focuses on this antibody subclass.

2.3. Parameter calibration and validation

To calibrate the parameters of the mathematical model and align its predictions with antibody concentration data from the cohort study, we employed Differential Evolution (DE), a bio-inspired stochastic optimisation algorithm [57]. This population-based method was used to estimate the unknown parameters of the model described in Section 2.2. Calibration was performed using a subset of the cohort data,

while the remaining data were reserved for independent validation. The configuration of the DE algorithm is summarised in Table 1.

Calibration requires defining an objective function that quantifies the discrepancy between the antibody levels predicted by the model and those measured in the cohort study. For this purpose, we used data from a cohort that received two vaccine doses administered with an 8–12 week interval. The objective function, denoted by ϵ , is defined in Eq. (14) as the minimum relative error between the experimental IgG antibody concentrations (IgG) [49] and the corresponding model predictions ($I\hat{g}G$):

$$\epsilon = \min (R_E(IgG, I\hat{g}G)). \quad (14)$$

The relative error R_E is expressed as:

$$R_E(IgG, I\hat{g}G) = \frac{\|IgG(t) - I\hat{g}G(t)\|_2}{\|IgG_{std}(t)\|_2}, \quad (15)$$

where $IgG_{std}(t)$ denotes the standard deviation of IgG measurements at each time point. We selected the L2 norm ($\|\cdot\|_2$) after testing alternatives, as it consistently produced the lowest R_E values and the closest agreement between simulations and experimental data.

Calibration was performed within physiologically motivated parameter ranges, summarised in Table 2. Parameters were sampled from uniform distributions across these ranges to ensure broad exploration of the search space. Candidate parameter sets with R_E below 5% were retained, yielding a pool of more than 16,000 feasible solutions. From this pool, the standard parameter set reported in Table 2 was obtained by averaging across all accepted candidates. This approach not only produced a representative set of parameters but also captured the variability and robustness of the calibration process.

Once calibrated, the model was validated under different experimental conditions, including two-dose and three-dose regimens. These tests demonstrated that the calibrated model is robust and adaptable, successfully reproducing the observed antibody dynamics under varying vaccination schedules.

The calibration problem addressed here corresponds to a single-objective, constrained optimisation in a continuous parameter space. The optimisation landscape is non-convex, potentially multimodal, and derivative-free, with moderately expensive evaluations since each function call requires solving a system of ODEs. These characteristics motivated our choice of DE, a robust population-based strategy. Although this work focuses on single-objective calibration, recent advances in evolutionary optimisation have introduced multi-objective and diversity-preserving methods with improved convergence and robustness [63–68]. While beyond the scope of the present study, such approaches could be incorporated in future work to balance competing objectives such as accuracy, robustness, and generalisability.

In addition to relative error, we also computed the Root Mean Square Error (RMSE), a complementary metric for quantifying model accuracy in absolute terms. Unlike R_E , which normalises deviations by measurement uncertainty, the RMSE directly measures the average magnitude of prediction errors in the same units as the experimental data (EU/ml):

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (IgG(t_i) - I\hat{g}G(t_i))^2}, \quad (16)$$

where $IgG(t_i)$ are the observed antibody levels at time t_i , $I\hat{g}G(t_i)$ are the corresponding model predictions, and n is the total number of observations. Using both R_E and RMSE provides complementary perspectives on model performance: the first emphasises consistency with measurement uncertainty, while the second captures the typical size of deviations in biologically interpretable units.

Table 1
Differential Evolution (DE) configuration for model calibration.

Component	Description
Objective function	Minimisation of the weighted relative error between model-predicted IgG antibody concentrations and clinical cohort data across five vaccination scenarios. Calibration was restricted to the nine most sensitive parameters.
DE strategy	best1bin strategy, in which new candidate solutions are generated by perturbing the best individual of the current population, promoting efficient convergence.
Population size (N_p)	$N_p = 100$, selected to balance exploration of the parameter space with computational cost. This size maintains sufficient diversity to avoid premature convergence while keeping runtime manageable.
Stopping criterion	Optimisation terminates after a fixed number of generations, with the maximum set to $\text{maxiter} = 10$.
Parameter bounds	Search restricted to physiologically plausible lower and upper bounds for each of the nine calibrated parameters (see Table 2).

Table 2
Baseline values, calibrated parameters, and initial values of the model.

Parameter	Interval $\rightarrow$ Calibrated value	Unit
δ_v	$(1.0 \times 10^{-4}, 1.0) \rightarrow 2.00 \times 10^{-2}$	(day ⁻¹)
$\delta_{v_{\text{res}}}$	$(1.0 \times 10^{-4}, 1.0) \rightarrow 6.50 \times 10^{-2}$	[day ⁻¹ (EU/ml ⁻¹)]
α_s	$(1.0 \times 10^0, 1.0 \times 10^4) \rightarrow 4.40 \times 10^{+1}$	(day ⁻¹)
$\delta_{s_{\text{res}}}$	$(1.0 \times 10^{-4}, 1.0) \rightarrow 4.08 \times 10^{-1}$	[day ⁻¹ (EU/ml ⁻¹)]
$\delta_{s_{\text{ap}}}$	$(1.0 \times 10^{-4}, 1.0) \rightarrow 2.00 \times 10^{-3}$	[day ⁻¹ (cells/ml ⁻¹)]
α_{ap}	$(1.0 \times 10^{-4}, 1.0) \rightarrow 2.90 \times 10^{-3}$	(day ⁻¹)
$\gamma_{s_{\text{ap}}}$	$(1.0 \times 10^{-9}, 1.0 \times 10^{-5}) \rightarrow 5.90 \times 10^{-7}$	[day ⁻¹ (spike/ml ⁻¹)]
δ_{apm}	$(1.0 \times 10^{-4}, 1.0) \rightarrow 1.10 \times 10^{-1}$	(day ⁻¹)
α_{th}	$(1.0 \times 10^{-4}, 1.0) \rightarrow 4.34 \times 10^{-3}$	(day ⁻¹)
β_{th}	$(1.0 \times 10^{-9}, 1.0 \times 10^{-5}) \rightarrow 4.64 \times 10^{-6}$	[day ⁻¹ (cells/ml ⁻¹)]
π_{th}	$(1.0 \times 10^{-7}, 1.0 \times 10^{-3}) \rightarrow 1.00 \times 10^{-5}$	[day ⁻¹ (cells/ml ⁻¹)]
δ_{the}	$(1.0 \times 10^{-4}, 1.0) \rightarrow 3.00 \times 10^{-1}$	(day ⁻¹)
α_{bn}	$(1.0 \times 10^{-4}, 1.0) \rightarrow 4.30 \times 10^{-3}$	(day ⁻¹)
β_{bn}	$(1.0 \times 10^{-8}, 1.0 \times 10^{-3}) \rightarrow 6.00 \times 10^{-6}$	[day ⁻¹ (cells/ml ⁻¹)]
δ_{bn}	$(1.0 \times 10^{-5}, 1.0 \times 10^{-1}) \rightarrow 1.00 \times 10^{-4}$	(day ⁻¹)
γ_{bn}	$(1.0 \times 10^{-6}, 1.0 \times 10^{-2}) \rightarrow 2.85 \times 10^{-4}$	[day ⁻¹ (spike/ml ⁻¹)]
$\beta_{ba_{pc}}$	$(1.0 \times 10^{-4}, 1.0) \rightarrow 4.00 \times 10^{-3}$	[day ⁻¹ (cells/ml ⁻¹)]
δ_{ba}	$(1.0 \times 10^{-4}, 1.0) \rightarrow 2.00 \times 10^{-1}$	(day ⁻¹)
β_{bm}	$(1.0 \times 10^{-7}, 1.0 \times 10^{-3}) \rightarrow 1.48 \times 10^{-5}$	[day ⁻¹ (cells/ml ⁻¹)]
k_0	$(1.0, 1.0 \times 10^2) \rightarrow 7.69$	dimensionless
k_1	$(1.0 \times 10^{-8}, 1.0 \times 10^{-3}) \rightarrow 2.44 \times 10^{-5}$	(ml/cells)
B_{max}	$(1.0 \times 10^5, 1.0 \times 10^8) \rightarrow 1.00 \times 10^7$	(cells/ml)
$\text{boost}_{bm_{pc}}$	$(1.0 \times 10^{-5}, 1.0 \times 10^{-2}) \rightarrow 4.00 \times 10^{-3}$	[day ⁻¹ (spike/ml ⁻¹)]
$\text{pool}_{bm_{pc}}$	$(1.0 \times 10^{-7}, 1.0 \times 10^{-3}) \rightarrow 2.00 \times 10^{-5}$	(day ⁻¹)
δ_{bm}	$(1.0 \times 10^{-5}, 1.0 \times 10^{-2}) \rightarrow 3.02 \times 10^{-4}$	(day ⁻¹)
δ_{pc}	$(1.0 \times 10^{-4}, 1.0) \rightarrow 5.30 \times 10^{-3}$	(day ⁻¹)
π_{pc}	$(1.0 \times 10^{-6}, 1.0 \times 10^{-2}) \rightarrow 2.00 \times 10^{-4}$	[day ⁻¹ (EU/ml ⁻¹)]
δ_{igg}	$(1.0 \times 10^{-4}, 1.0) \rightarrow 2.31 \times 10^{-1}$	(day ⁻¹)
Initial values		
Variable	Description	Initial value
V	SARS-CoV-2	1.35×10^6 copies/ml
S	Spike protein	0 particles
A_p	Naïve presenting cells	1.00×10^6 cells/ml
A_{pm}	Mature presenting cells	0 cells/ml
T_{bn}	Naïve CD4+ T cells	1.00×10^6 cells/ml
T_{he}	Effector CD4+ T cells	0 cells/ml
B_n	Naïve B cells	2.5×10^5 cells/ml
B_a	Active B cells	0 cells/ml
B_m	Memory B cells	0 cells/ml
P_c	Plasma cells	0 cells/ml
I_{gg}	IgG antibodies	0 EU/ml

2.3.1. Literature support for model parameters

The calibrated value $\delta_v = 0.02 \text{ day}^{-1}$ for virus clearance is at the low end of reported free-virus clearance rates. For example, hepatitis B virus clearance has been estimated between ~ 0.69 and 21.7 day^{-1} [69]. This wide range suggests that very slow viral decay ($\approx 0.02 \text{ day}^{-1}$) lies within plausible bounds, especially if virus persists in protected reservoirs. Neutralising antibodies accelerate viral clearance. In one SARS-CoV-2 model, the antibody neutralisation constant was $k_{Ab} \approx 0.11 \text{ mL}/(\text{unit day})$ [70]. Converting units, this is of the same order as $\delta_{v_{\text{igg}}} = 0.065 \text{ day}^{-1}/(\text{EU/ml})$, supporting our fitted value. Direct literature estimates for spike-protein production (α_s) or IgG-mediated spike clearance ($\delta_{s_{\text{igg}}}$) are scarce. Antibody neutralisation constants

($\sim 0.1 \text{ mL}/\text{U/day}$ [70]) imply that bound spike is cleared on a comparable timescale. Rates of antigen uptake by APCs are less documented, but phagocytic clearance is generally slower than free-virus clearance. Our fitted values fall within expected biological orders of magnitude.

Models of immune dynamics often use APC turnover rates on the order of $0.1\text{--}1.0 \text{ day}^{-1}$. For instance, a COVID-19 vaccine model used APC renewal $\gamma_{APC} \approx 0.49 \text{ day}^{-1}$ and APC death $\delta_{APC} \approx 0.20 \text{ day}^{-1}$ [70]. In comparison, our $\alpha_{ap} = 0.0029 \text{ day}^{-1}$ and $\delta_{apm} = 0.11 \text{ day}^{-1}$ are smaller but biologically reasonable. They imply slow APC activation and an APC lifespan of about 6–7 days, consistent with literature [70].

Naïve CD4⁺ T cells proliferate very slowly. Labelling studies estimate that only $\sim 0.2\%$ of naïve CD4⁺ T cells divide per day in young adults [71], implying a per-cell proliferation rate of $\approx 0.002 \text{ day}^{-1}$. Our $\alpha_{th} = 4.34 \times 10^{-3} \text{ day}^{-1}$ is of the same order. Peripheral loss rates of naïve CD4⁺ cells have been estimated at $\delta \approx 0.0025 \text{ day}^{-1}$ [71].

Effector CD4⁺ T cells are short-lived once antigen is cleared. Models report effector death rates of ~ 0.0075 to 0.39 day^{-1} [70]. Our $\delta_{the} = 0.30 \text{ day}^{-1}$ (half-life ≈ 2.3 days) lies within this range. T-cell stimulation parameters (β_{th} , π_{th}) govern contact-induced activation and proliferation. We did not find direct estimates, but fitted values of $10^{-6}\text{--}10^{-5}$ (per cell day) imply very rare productive encounters, which is plausible given lymphocyte densities.

Naïve B cells divide extremely slowly. In vivo labelling studies show that naïve B cells divide at only $\sim 0.46\%$ per day ($\approx 0.0046 \text{ day}^{-1}$) [72]. This closely matches our $\alpha_{bn} = 4.30 \times 10^{-3} \text{ day}^{-1}$. Naïve B-cell death is very slow. Our $\delta_{bn} = 1.0 \times 10^{-4} \text{ day}^{-1}$ corresponds to a half-life of many years, consistent with naïve B-cell longevity. Differentiation of activated B cells into plasma cells has been modelled at $\sim 0.1\text{--}1 \text{ day}^{-1}$ [70]. Our $\beta_{ba_{pc}} = 4.0 \times 10^{-3}$ is lower, reflecting strong T-cell dependence.

Plasma-cell death rates vary widely, from short-lived plasmablasts (days) to long-lived plasma cells (years). A model cited values from $\delta_p \approx 0.0083$ to 1.23 day^{-1} [70]. Our $\delta_{ba} = 0.20 \text{ day}^{-1}$ (half-life $\approx 3\text{--}4$ days) is consistent with short-lived antibody-secreting cells.

2.4. Sensitivity analysis and uncertainty quantification

Sensitivity analysis quantifies the extent to which variations in model parameters affect outputs, thereby identifying the most influential determinants of system behaviour. This analysis informs both model optimisation and biological interpretation by revealing which parameters require careful calibration or carry the largest predictive weight.

We employed the Python library SALib [73], which provides several global and local sensitivity analysis methods. Among these, Sobol' indices [74–76] were selected to capture both direct and interaction effects of parameters on model variance. First-order indices (S_i) measure the isolated contribution of a parameter, whereas total-order indices (S_{Ti}) quantify its overall impact, including higher-order interactions. In additive models, the sum of first-order indices equals one, but for non-additive systems such as our ODE-based formulation, this sum is typically less than one, while the sum of total-order indices is always greater than or equal to one [75,77].

Two complementary sensitivity analyses were performed: one considering all 31 parameters and another restricted to the four most

influential ones (B_{n0} , δ_{pc} , π_{pc} , and δ_{lgG}). Both analyses focused on antibody production dynamics, guided by clinical data [49,50,78]. To highlight parameters relevant to early immune activation, particularly after the first vaccine dose, we used an empirical Sobol index threshold of 0.07 (Fig. 5).

Because parameter values in vaccine immunology are often uncertain or incompletely reported, we complemented sensitivity analysis with uncertainty quantification (UQ). UQ was implemented using the Chaospy library [79], which propagates input uncertainty via polynomial chaos expansions with orthogonal bases (e.g., Hermite, Legendre, Chebyshev). Parameter uncertainties were modelled as normally distributed [80,81], with means equal to fitted values (FV) and standard deviations defined to ensure that 99.7% of samples lie within $\pm 25\%$ of FV :

$$N(\mu, \sigma) = N(FV, \frac{FV \times 0.25}{3}).$$

For example, if $FV = 100$, then $\sigma \approx 8.33$, producing a plausible range of 75–125. This formulation leverages the Empirical Rule (68–95–99.7 rule) [82] and the Bienaymé–Chebyshev inequality [83, 84] to maintain biological plausibility without overstating precision.

Together, sensitivity analysis and UQ provide a robust framework for identifying critical parameters, quantifying the effects of variability, and improving confidence in model-based predictions of antibody dynamics.

2.5. Computational implementation

The model was implemented in Python using the `solve_ivp` function from the `scipy.integrate` package [85]. Numerical integration was performed with the LSODA method, which automatically switches between non-stiff and stiff solvers depending on the system characteristics. Differential Evolution (DE) was implemented in Python with the NumPy library (version 1.24.3).

All simulations were executed in Python 3.11.3 (64-bit) on a workstation equipped with an Intel® Core™ i5-9300HF processor, 16 GB RAM, and Windows 11 Home (version 22H2).

Computational cost was not a limiting factor: a single model run required only a few seconds, while complete DE optimisation and sensitivity analyses finished within two hours.

Table 2 summarises the model's initial conditions, population acronyms, and parameter definitions.

2.6. Dose optimisation

We performed 388 simulations to evaluate how the timing of the second dose of ChAdOx1 nCoV-19 influences antibody production and the duration of protection. Simulations were initialised on day 63, consistent with the 8–12 week cohort in clinical studies [49,50], and extended to day 450. Protection was defined as the number of days during which IgG levels exceeded the protective threshold of 218 EU/ml [78].

To optimise long-term immunity, we extended simulations to 500 days and analysed two vaccination strategies: a two-dose and a three-dose regimen. For each scenario, we calculated the Percentage of Protected Days (PPD), defined as the proportion of the 500-day window during which IgG remained above the protective threshold. Sensitivity analyses were also conducted using lower (163.5 EU/ml) and higher (272.5 EU/ml) thresholds.

In the two-dose scenario, the second dose was varied daily between day 28 and day 450. In the three-dose scenario, the second dose was varied between day 28 and day 225, while the third dose was systematically tested starting 28 days after the second and extending to day 450.

3. Results

This section presents a detailed comparison between predictions of our computational model and data from cohorts vaccinated with the ChAdOx1 nCoV-19 (AZD1222) vaccine [49,50,78]. We also evaluate model sensitivity, quantify prediction uncertainties, and identify vaccination schedules that maximise protection over a 500-day period following the first dose.

3.1. Creating and validating a virtual patient cohort

Fig. 2 summarises the immune cell interactions involved in the response to ChAdOx1 nCoV-19, while Fig. 3 depicts the simulated dynamics under a two-dose schedule, with the second dose at day 63.

After administration, the vaccine delivers a modified adenovirus encoding the SARS-CoV-2 spike protein. Naïve antigen-presenting cells (A_p) capture the viral antigen and mature into active antigen-presenting cells (A_{pm}). As shown in Fig. 3, this process reduces the naïve population while increasing the mature one.

Mature antigen-presenting cells activate helper T lymphocytes (T_{hn}), which differentiate into effector cells (T_{he}). Effector cells then stimulate B lymphocytes, driving their transition from naïve (B_n) to active (B_a) states. This cascade produces plasma cells (P_c) and antibodies, reflected in the IgG increase in Fig. 3. Activated B cells also generate memory cells (B_m), ensuring long-term immunity.

As discussed, the model integrates key biological processes, homeostasis, self-replication, neutralisation, and antigen–antibody complex formation. These mechanisms are quantitatively represented in Fig. 3, providing insight into immune population dynamics across vaccination.

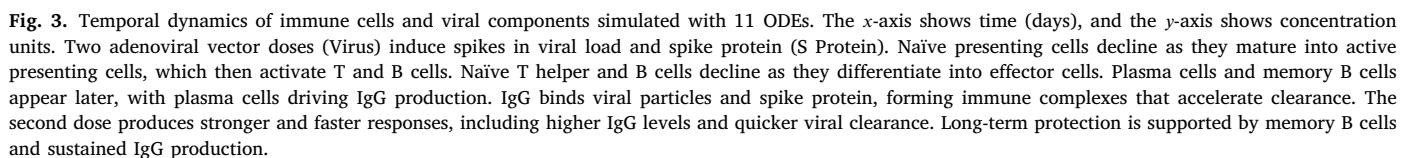
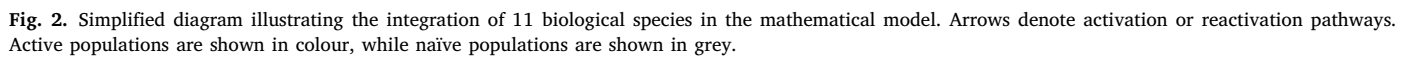
The framework is implemented as a system of 11 ordinary differential equations (ODEs) with 31 parameters, described in Section 2. Agreement between the conceptual diagram (Fig. 2) and numerical results (Fig. 3) confirms that the model reproduces central immunological processes.

We performed five numerical experiments to simulate antibody responses to ChAdOx1 nCoV-19. Four tested two-dose regimens with intervals of 8–12 weeks, 15–25 weeks, and 44–45 weeks, while the fifth evaluated a three-dose regimen. In the latter, two initial doses were given 8–16 weeks apart, followed by a third after 30 weeks. Cohort data were drawn from published studies [49,50,78]. Blood samples were available for 115 participants (8–12 week group), 116 (15–25 weeks), and 30 (44–45 weeks). Antibody persistence one year after the second dose was confirmed in 704 participants from the 8–12 week group [50]. The three-dose trial included 80 participants. Details are provided in Section 2.

A prior study associated a median antibody level of 218 ELISA Units/ml (EU/ml) at day 28 with 50% viral neutralisation (PRNT50) [78]. This protective threshold is indicated in all antibody response figures.

In the first experiment, shown in Fig. 4, the model was calibrated using six data points from the 8–12 week cohort. Parameters were optimised with Differential Evolution (DE) [57], minimising relative error between observed and simulated data. Details of calibration are in Section 2.3. DE produced 16,000 parameter sets with relative errors below 5%, defining the virtual cohort (VC). For subsequent experiments, the same parameters were retained, with only booster timing varied (days 114, 313, 69, and 98/308).

Model validation was based on relative error between simulated means and cohort data [49,50] (Fig. 4). Two-dose regimens yielded errors of 3.01% (8–12 weeks), 0.42% (one-year cohort), 0.59% (15–25 weeks), and 0.13% (44–45 weeks). The three-dose regimen had 1.84% error. These results indicate that longer intervals between the first and second doses improve alignment with observed immune responses. Further error analyses are presented in Section 2.3 (see Table 3).



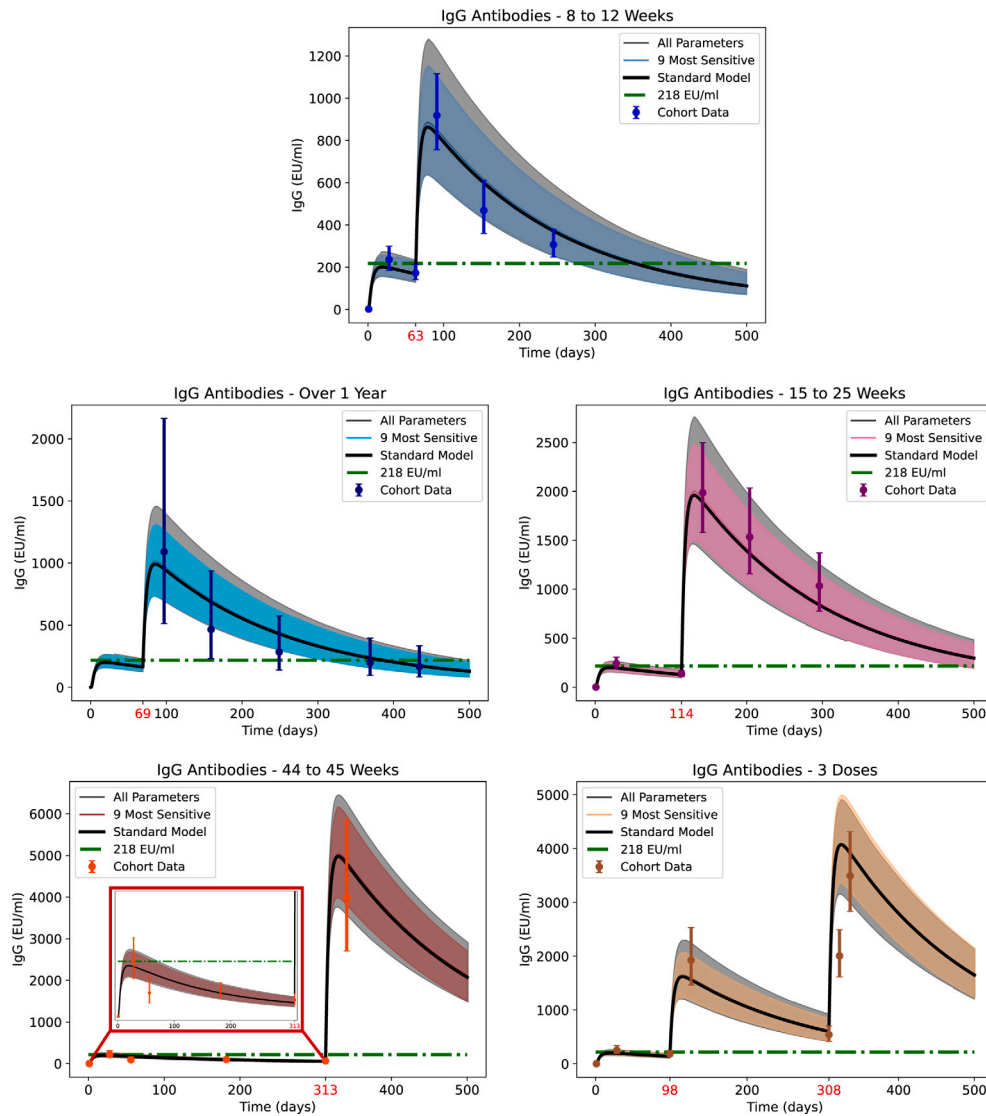


Fig. 4. Comparison of observed antibody levels [49,50] (dots with IQR error bars) and model simulations (solid lines). Second doses occurred at days 63 (8–12 weeks), 69 (one-year cohort), 114 (15–25 weeks), and 313 (44–45 weeks). In the three-dose regimen, doses were on days 0, 98, and 308. The protective threshold (218 EU/ml) is indicated by the green dash-dotted line. Uncertainty quantification shows parameter variation effects: all 31 parameters varied (grey) and the four most sensitive parameters varied (coloured), both at 25%.

Table 3

Relative errors and RMSE values comparing model simulations and cohort IgG data across vaccination scenarios.

Scenario	Relative error (%)	RMSE (EU/ml)
8–12 weeks (2 doses)	3.01	71.60
Over 1 year (2 doses)	0.42	135.14
15–25 weeks (2 doses)	0.59	114.10
44–45 weeks (2 doses)	0.13	333.98
Three-dose regimen	1.84	804.47

3.2. Identifying and varying the model's most sensitive parameters

Fig. 4 shows model outputs under parameter variation, with shaded regions representing uncertainty quantification (UQ). Two UQ approaches are presented: (a) full UQ, varying all 31 model parameters, and (b) reduced UQ, varying only those identified as most influential through sensitivity analysis. Sobol indices were applied to quantify each parameter's contribution to output variance, enabling the identification of parameters with the greatest impact on model predictions. These variations both contribute to predictive uncertainty

and form the basis of the Virtual Cohort (VC), capturing realistic inter-individual variability. Fig. 5 displays first-order Sobol indices for all vaccination regimens, highlighting parameters most strongly associated with antibody production.

Sensitivity analysis identified nine key parameters influencing IgG dynamics: the viral spike protein release rate (δ_s), spike protein neutralisation by antigen-presenting cells ($\delta_{s_{ap}}$), APC maturation rate upon spike exposure ($\gamma_{s_{ap}}$), baseline naïve B cell population (B_{n0}), memory B cell growth parameters (k_0 , k_1), plasma cell decay rate (δ_{pc}), antibody production rate by plasma cells (π_{pc}), and IgG decay rate (δ_{IgG}). Each parameter was modelled as normally distributed, with standard deviations derived using the empirical rule [86], ensuring 99.7% coverage within a $\pm 25\%$ range of the mean. Methodological details are provided in Section 2.4. Although five dosing schedules were simulated, the three-dose regimen (Fig. 4) captures the representative sensitivity patterns of all influential parameters. Additional results are included in the Supplementary Material.

Fig. 5 highlights parameters exceeding a Sobol index of 0.07, underscoring their influence across the 500-day simulation. Four parameters exhibit consistently high sensitivity, while others act in a phase-specific manner. The legend details their immunological relevance.

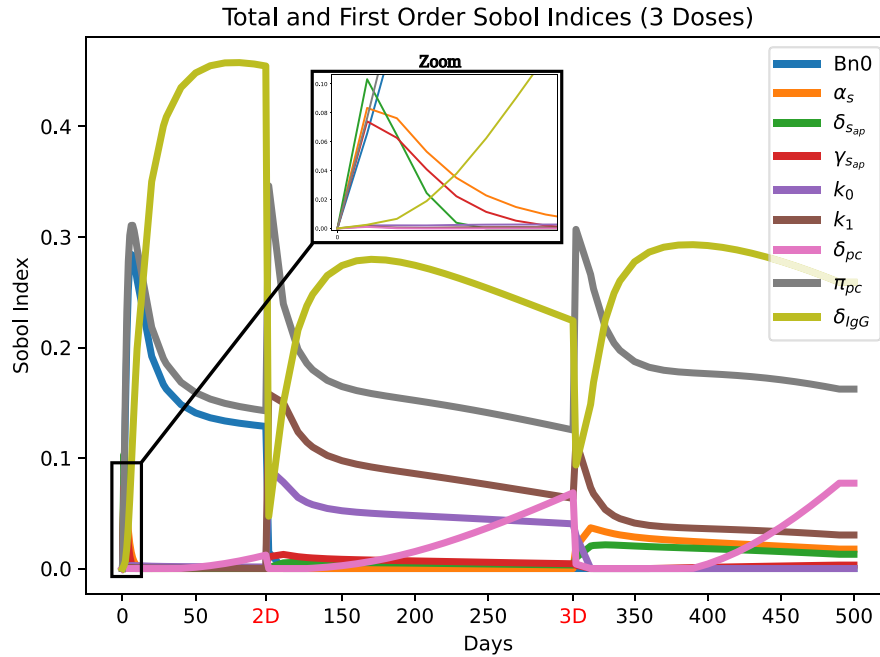


Fig. 5. First-order Sobol indices for the three-dose ChAdOx1 nCoV-19 regimen. Red markers on the x-axis indicate the second (day 98) and third (day 308) doses. Solid lines show time-varying sensitivity. Only parameters exceeding a Sobol index of 0.07 are displayed. The IgG decay rate (δ_{IgG} , yellow) drops sharply in sensitivity following each dose, while the IgG production rate (π_{pc} , grey) gains influence post-booster. The naïve B cell population (B_{n0} , blue) dominates early sensitivity, whereas memory B cell replication (k_1 , brown) becomes more influential later. The inset zooms into early dynamics.

Phase-dependent sensitivity was also observed in additional parameters. The spike protein production rate (α_s , orange), its presentation by APCs (δ_{sap} , green), and APC maturation (γ_{sap} , red) primarily influenced early stages before B and plasma cell activation. In contrast, plasma cell decay (δ_{pc} , pink) played a greater role in long-term antibody decline.

Overall, the sensitivity analysis reveals stage-dependent influences, with B cell and antibody-related processes emerging as dominant regulators of vaccine-induced immunity. These findings emphasise the complexity of immune dynamics and inform parameter selection for robust simulation design.

3.3. Dose optimisation

Fig. 6 summarises the optimisation results. In the two-dose regimen, the model identified day 94 (week 13) as the optimal timing for the second dose, providing protection for approximately 81% of the 500-day period. A broader interval between weeks 11 and 21 still maintained over 70% protection, offering flexibility under supply or logistical constraints.

For the three-dose regimen, the optimal schedule involved a second dose on day 30 (week 4) and a third on day 117 (week 16). This schedule yielded protection for 93.8% of the 500-day period, reflecting enhanced and sustained immunity compared with the two-dose regimen.

The threshold sensitivity analysis (Fig. 6), which explored a $\pm 25\%$ variation around the 218 EU/ml baseline, revealed that lowering the threshold (163.5 EU/ml) shifted optimal timings earlier and increased PPD (98%), while raising it (272.5 EU/ml) delayed optimal timings and reduced PPD (92%). Across all scenarios, the three-dose regimen consistently outperformed the two-dose regimen in maintaining long-term protection.

4. Discussion

Our mathematical model, described in Section 2, successfully reproduced the immune response to the ChAdOx1 nCoV-19 vaccine across diverse dosing regimens. Predictions closely matched observed

antibody levels from clinical cohort data [49], as shown in Fig. 4, without requiring scenario-specific parameter adjustments. This consistency underscores the model's generalisability across different vaccination schedules.

The Virtual Cohort (VC) used in this study was anchored to a reference population corresponding to the empirical IgG data represented by the blue curve in the top panel of Fig. 4. This group, consisting of 115 individuals [49], served as the calibration basis for reproducing realistic antibody dynamics following vaccination.

The VC comprised over 16,000 virtual patients, generated by sampling parameter sets that reproduced the empirical IgG trajectory. Only simulations yielding a normalised error below 5% were retained. Parameter optimisation was performed using the Differential Evolution algorithm, which efficiently explored the parameter space to identify biologically plausible configurations consistent with observed immune responses.

By anchoring the VC to experimentally validated data and applying strict selection criteria, the model ensured that virtual predictions remained grounded in real-world variability. This robust virtual population enabled reliable sensitivity analyses and simulations of diverse vaccination scenarios.

Additional trajectories in Fig. 4, representing alternative dosing strategies, further validated the model's ability to generalise beyond the reference scenario. In two-dose regimens, simulated IgG dynamics closely matched observed kinetics, capturing both peak amplitude and decay trends [49]. This alignment suggests that the model accurately represents the immunological processes underlying vaccine response.

Simulations also confirmed the persistence of IgG beyond one year post-vaccination (yellow curve in Fig. 4), consistent with clinical observations [50]. Individuals who received two standard doses 69 days apart maintained detectable antibody levels for more than a year, supporting sustained protection against severe disease.

For three-dose regimens, the model reproduced the booster-induced rise in IgG levels, including increased peak magnitude and extended antibody persistence [49]. These results underscore the model's ability to capture adaptive immune memory and secondary responses. Overall,

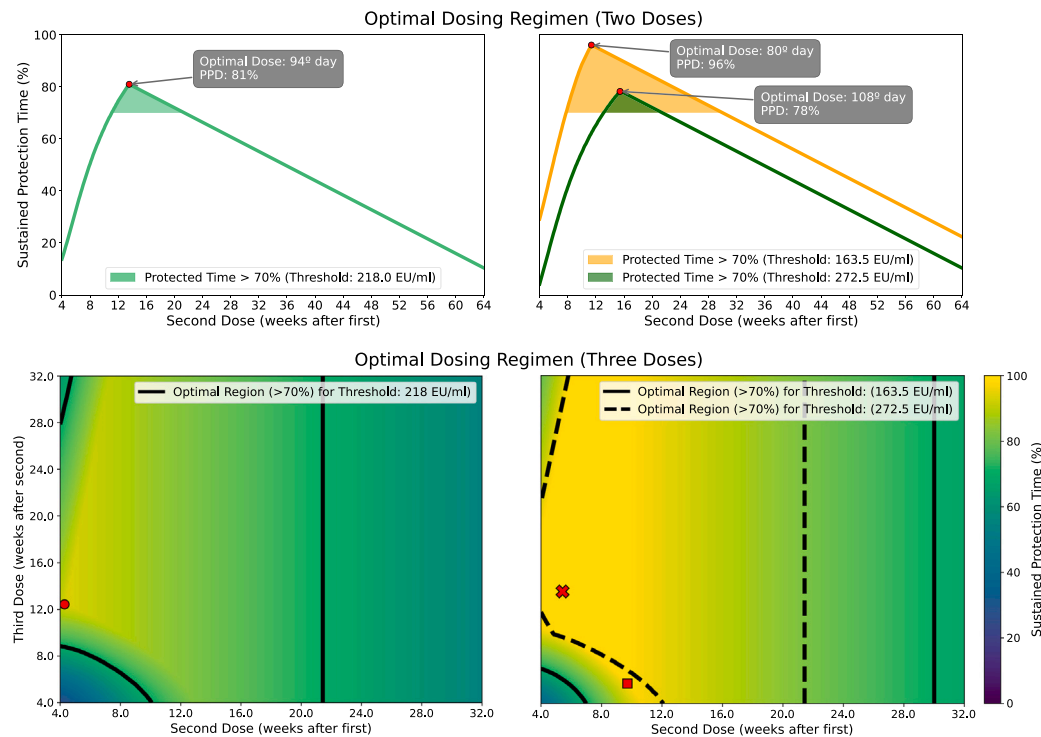


Fig. 6. Simulation results over 500 days showing optimal dosing schedules. **Top Left:** Two-dose regimen with threshold 218 EU/ml, maximum protection of 81% (PPD) at day 94. **Top Right:** Sensitivity analysis with lower (163.5 EU/ml) and higher (272.5 EU/ml) thresholds, optimal timings at days 80 (96% PPD) and 108 (78% PPD), respectively. **Bottom Left:** Three-dose regimen (threshold 218 EU/ml), maximum protection of 94% (PPD) with the second dose at week 4 and the third at week 12; the red dot indicates the peak response, and the solid black line delimits the protected region (>70% above threshold). **Bottom Right:** Sensitivity analysis of three doses: at the lower threshold (163.5 EU/ml), optimal protection (98% PPD) occurs with the second dose at week 10 and the third at week 5 after the second (red square); at the higher threshold (272.5 EU/ml), optimal protection (92% PPD) occurs with the second dose at week 5 and the third at week 13 after the second (red cross). Solid and dashed lines indicate protected regions for the two thresholds.

the close agreement between simulations and empirical data supports the reliability of both the model and the VC framework.

Simulations of dose intervals (Fig. 4) revealed a non-linear relationship between spacing and immune protection: longer intervals enhanced peak titres but did not proportionally extend protection. This outcome reflects the role of memory B-cell dynamics, constrained by a replication limit ($B_{m_{\max}}$), which enables stronger responses to delayed boosters without substantially prolonging immunity duration (plateauing at 550–560 days post-booster).

Uncertainty quantification (Fig. 4) reinforced these findings, consistently showing elevated antibody levels with extended intervals. Sobol sensitivity analysis identified three dominant parameters: naïve B-cell homeostasis (B_{n_0}), plasma cell decay rate (δ_{pc}), and IgG synthesis/clearance rates (π_{pc} , δ_{lgG}). Temporal analysis revealed dynamic shifts in parameter influence: B_{n_0} governed early post-vaccination responses, whereas δ_{pc} became dominant after boosters, modulating long-term antibody maintenance.

The IgG decay rate (δ_{lgG}) also exhibited time-varying sensitivity. As shown in Fig. 5, its influence transiently decreased after the second dose, peaked, and then declined, with prolonged dose intervals amplifying its impact. The small difference between total and first-order Sobol indices justified the exclusive presentation of first-order effects.

The optimisation simulations identified optimal dose intervals based on a protection threshold of 218 EU/ml. Under this condition, a two-dose regimen with a 106-day interval achieved 81% protection time over 500 days (Fig. 6). For the three-dose regimen, intervals of 30 and 94 days yielded 94% protection time (Fig. 6). These specific outcomes, however, are dependent on the chosen threshold, as demonstrated in the previously reported sensitivity analysis where a $\pm 25\%$ variation produced a distinct set of optimal intervals.

We emphasise that the association between antibody titres and clinical protection is context-dependent and multifactorial. While antibody

kinetics are a valuable correlate of immunity, they are not a comprehensive surrogate for vaccine efficacy; cellular immunity, mucosal responses, and population-level dynamics also contribute materially to protection. Therefore, the candidate dosing schedules identified here should be regarded as mechanistically informed hypotheses that require further evaluation via population-level modelling, and experimental studies that include cellular readouts.

Furthermore, translating the optimal dosing schedules reported here into public-health policy requires consideration of logistical, epidemiological, and regulatory constraints. Examples include limited vaccine supply (which may favour strategies prioritising broader first-dose coverage over delayed boosters), variant circulation with differing immune-escape properties (necessitating variant-specific vaccine updates or shorter booster intervals), cold-chain and clinic capacity constraints, and population-level adherence to multi-dose schedules. The optimisation framework presented is modular and can incorporate such constraints: for instance, adding supply constraints or a deployment-delay penalty to the objective function enables optimisation under limited availability; scenario-based robust optimisation across multiple variant parameterisations can produce schedules that perform consistently under epidemic uncertainty. We emphasise that our schedules represent immunologically informed candidates whose adoption would require integration with logistical planning and population-level epidemiological modelling.

Despite these promising results, our study has several limitations. First, the model does not account for prior SARS-CoV-2 infection, focusing exclusively on vaccine-induced immunity, which limits its applicability to previously exposed populations. Second, the model primarily examines humoral immunity, specifically IgG responses, without incorporating the critical contributions of cellular immunity components such as CD8⁺ T cells and NK cells. Third, the simulated

vaccination schedules were aligned with clinical trial designs, with the second dose fixed at day 63, restricting assessment of earlier dosing intervals implemented in real-world practice. Fourth, model parameters were assumed to follow normal distributions with a $\pm 25\%$ range around calibrated values. This assumption does not fully capture variability across age groups, sexes, or individuals with comorbidities, and therefore likely underestimates biological heterogeneity. Fifth, our analysis focused exclusively on immune responses to the spike protein of the ancestral SARS-CoV-2 strain. We did not explicitly consider altered immunodynamics or immune escape associated with emerging variants, which limits the direct applicability of our predictions to the current epidemiological landscape. These aspects underscore the exploratory nature of the study and delineate key areas for future model refinement. We also acknowledge that approximating missing variances by the method adopted in this work [61] is a simplification and that a fully Bayesian inference of observational variances would be more rigorous. We plan to carry out the full Bayesian re-calibration and posterior predictive checks in follow-up work; until then, readers should interpret uncertainty bands presented here as conditional on the stated dispersion approximation. Finally, the clinical dataset used to calibrate model parameters in this study comprises adults aged 18–55. Consequently, direct extrapolation of the quantitative predictions to elderly (age > 65) or immunocompromised populations is limited. Ageing and immunocompromised conditions are associated with several immunobiological changes that can affect vaccine dynamics and the ensuing humoral response, including reduced naïve T- and B-cell pools, decreased proliferative capacity, altered cytokine milieu, slower seroconversion, and different antibody decay kinetics (immunosenescence). In practical terms, these differences may manifest, for example, as lower baseline homeostatic levels and altered effective rates of activation and clearance. Therefore, the optimised dosing schedules reported here should be considered applicable primarily to populations similar to the calibration cohort. For elderly or immunocompromised groups, we recommend recalibration of a subset of key parameters (for example, baseline lymphocyte levels, activation rates, and antibody decay rates) and targeted sensitivity analyses to establish robustness and to re-run the schedule optimisation under the recalibrated parameter distributions.

Despite these simplifications, the model robustly captures population-level immune patterns, offering a practical framework for evaluating vaccination strategies and optimising dosing schedules. In particular, it provides three key translational applications: establishing a quantitative foundation for booster scheduling; enabling exploration of fractional dosing strategies (e.g., 50% viral load regimens); and supporting the development of virtual patient cohorts for vaccine studies.

The model's predictive accuracy, supported by clinical validation [87,88], positions it as a versatile tool for optimising COVID-19 and broader vaccine platforms. Its advancements will help pave the way towards personalised vaccination regimens tailored to individual immune profiles and evolving viral landscapes.

Future research will address several key issues. First, although the present model does not explicitly represent CD8⁺ T cells or other cytotoxic lymphocytes, their functional contribution is partly embedded in parameters that control viral clearance and antigen decay. In this way, the antiviral activity of cytotoxic T cells is indirectly reflected within the fitted dynamics. Nevertheless, this implicit representation cannot fully capture the complexity of cellular immunity, such as memory T-cell formation or variant cross-reactivity. We therefore acknowledge this simplification as a limitation of the current study and identify the explicit integration of CD8⁺ T cells and other cellular components as a priority for future model refinement. We believe that integrating cellular immunity dynamics, particularly T-cell responses, will enable a more comprehensive understanding of long-term protection, especially in reinfection scenarios or heterogeneous immune backgrounds. This will require the acquisition of new data, such as post-vaccination T-cell

counts. Second, the development of stochastic models will improve the representation of biological variability, allowing assessment of seroconversion failure and identification of individuals at risk of insufficient immune responses, considering factors such as comorbidities, vaccination timing, and viral load. Third, the model will be expanded to analyse variant-specific immune responses, providing critical insights into how mutations affect protection and facilitating the development of targeted strategies against emerging variants. Finally, we note that alternative multi-objective or decomposition-based metaheuristics (such as MOGMO, MOEDO, and decomposition-based MOEAs) are promising and could be benchmarked against DE in future work; however, DE's simplicity, reproducibility, and parallelisability made it an appropriate and practical choice for the present calibration task.

We summarise potential weaknesses and sources of bias to guide interpretation and future work:

- **Data quality and completeness:** clinical datasets used for calibration occasionally lacked dispersion measures, which we approximated; missing metadata (e.g. comorbidities, age distributions) limit cohort stratification.
- **Population representativeness:** calibration was performed on cohorts aged 18–55; extrapolation to elderly or immunocompromised populations is therefore limited without recalibration.
- **Model structural omissions:** the present model focuses on humoral dynamics and simplifies or omits some cellular processes (CD8⁺ responses, innate cell signalling), which can affect antigen clearance and long-term protection.
- **Parameter uncertainty and identifiability:** several parameters are weakly constrained by available data, which can broaden predictive intervals and lead to parameter combinations that trade off mechanistic interpretations.
- **Threshold-based conclusions:** optimisation results that rely on a single protective threshold are conditional on that threshold.

CRediT authorship contribution statement

Maicom Peters Xavier: Writing – review & editing, Writing – original draft, Validation, Resources, Methodology, Formal analysis, Data curation, Conceptualization. **Rodrigo Weber dos Santos:** Writing – review & editing, Validation, Supervision, Methodology, Formal analysis, Conceptualization. **Marcelo Lobosco:** Writing – review & editing, Validation, Supervision, Methodology, Formal analysis, Data curation, Conceptualization.

Code availability

The source code of the implementation is available in the following GIT: <https://github.com/MaicomPX/ode-modeling-chadox1-immunity>.

Ethics statement

This study did not involve human participants or animal experiments. The analyses were performed using a mathematical model and publicly available aggregated clinical data. Therefore, ethical approval and informed consent were not required.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used ChatGPT in order to improve the readability and language of the manuscript. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Funding

This work has been supported by UFJF, by CAPES, Brazil - Finance Code 001; by CNPq, Brazil - Grant number 308745/2021-3; and by FAPEMIG, Brazil Grant number APQ 02830/17 and APQ-02513-22; by FINEP, Brazil SOS Equipamentos 2021 AV02 0062/22.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Maicom Peters Xavier reports financial support was provided by Federal University of Juiz de Fora. Maicom Peters Xavier reports financial support was provided by Coordination for the improvement of Higher Education Personnel. Maicom Peters Xavier reports financial support was provided by National Council for Scientific and Technological Development. Maicom Peters Xavier reports financial support was provided by Minas Gerais State Foundation of Support to the Research. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The authors would like to thank Carla Rezende Barbosa Bonin, Alexandre Bittencourt Pigozzo, and Bernardo Martins Rocha for their valuable comments and insightful suggestions that helped improve this paper.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.compbiomed.2025.111149>.

References

- [1] T.P. Velavan, C.G. Meyer, The COVID-19 epidemic, *Trop. Med. Int. Health* 25 (3) (2020) 278.
- [2] L. Yang, S. Liu, J. Liu, Z. Zhang, X. Wan, B. Huang, Y. Chen, Y. Zhang, COVID-19: immunopathogenesis and immunotherapeutics, *Signal Transduct. Target. Ther.* 5 (1) (2020) 128.
- [3] S. Feng, D.J. Phillips, T. White, H. Sayal, P.K. Aley, S. Bibi, C. Dold, M. Fuskova, S.C. Gilbert, I. Hirsch, et al., Correlates of protection against symptomatic and asymptomatic SARS-CoV-2 infection, *Nature Med.* 27 (11) (2021) 2032–2040.
- [4] C. DeLisi, J.A. Berzofsky, T-cell antigenic sites tend to be amphipathic structures., *Proc. Natl. Acad. Sci.* 82 (1985) 7048–7052.
- [5] A.S.D. Groot, A. Bosma, N. Chinai, J. Frost, B.M. Jesdale, M.A. Gonzalez, W. Martin, C. Saint-Aubin, From genome to vaccine: in silico predictions, ex vivo verification, *Vaccine* 19 (31) (2001) 4385–4395, [http://dx.doi.org/10.1016/S0264-410X\(01\)00145-1](http://dx.doi.org/10.1016/S0264-410X(01)00145-1).
- [6] N. Kumar, B.S. Hendriks, K.A. Janes, D. de Graaf, D.A. Lauffenburger, Applying computational modeling to drug discovery and development, *Drug Discov. Today* 11 (17–18) (2006) 806–811, <http://dx.doi.org/10.1016/j.drudis.2006.07.010>.
- [7] I. Doytchinova, D. Flower, Vaxijen: a server for prediction of protective antigens, tumour antigens and subunit vaccines, *BMC Bioinformatics* 8 (2007) <http://dx.doi.org/10.1186/1471-2105-8-4>.
- [8] M. Davies, J. Bayry, E. Tchilian, J. Vani, M. Shaila, E. Forbes, S. Draper, P. Beverley, D. Tough, D. Flower, Toward the discovery of vaccine adjuvants: coupling in silico screening and in vitro analysis of antagonist binding to human and mouse ccr4 receptors, *Plos One* 4 (2009) e8084, <http://dx.doi.org/10.1371/journal.pone.0008084>.
- [9] B.F. Haynes, G. Kelsoe, S.C. Harrison, T.B. Kepler, B-cell-lineage immunogen design in vaccine development with HIV-1 as a case study, *Nature Biotechnol.* 30 (5) (2012) 423–433, <http://dx.doi.org/10.1038/nbt.2197>.
- [10] A. Tyagi, P. Kapoor, R. Kumar, K. Chaudhary, A. Gautam, G. Raghava, In silico models for designing and discovering novel anticancer peptides, *Sci. Rep.* 3 (2013) <http://dx.doi.org/10.1038/srep02984>.
- [11] A. Daina, O. Michielin, V. Zoete, SwissADME: a free web tool to evaluate pharmacokinetics, drug-likeness and medicinal chemistry friendliness of small molecules, *Sci. Rep.* 7 (1) (2017) <http://dx.doi.org/10.1038/srep42717>.
- [12] F. Pappalardo, D. Flower, G. Russo, M. Pennisi, S. Motta, Computational modelling approaches to vaccinology, *Pharmacol. Res.* 92 (2015) 40–45.
- [13] E. Lee, D. Teuwen, A. Pirani, K. Gernert, B. Marzolf, K. Kennedy, D. Mobbs, R. VOOEAnco, A. Aderem, B. Pulendran, Systems biology approach predicts immunogenicity of the yellow fever vaccine in humans, *Nature Immunol.* 10 (2008) 116–125, <http://dx.doi.org/10.1038/ni.1688>.
- [14] D.R. Flower, I.K. Macdonald, K. Ramakrishnan, M.N. Davies, I.A. Doytchinova, Computer aided selection of candidate vaccine antigens, *Immunome Res.* 6 (Suppl 2) (2010) S1, <http://dx.doi.org/10.1186/1745-7580-6-s2-s1>.
- [15] A. Nandy, S. Basak, A brief review of computer-assisted approaches to rational design of peptide vaccines, *Int. J. Mol. Sci.* 17 (5) (2016) 666, <http://dx.doi.org/10.3390/ijms17050666>.
- [16] K.S. Allemailem, A comprehensive computer aided vaccine design approach to propose a multi-epitopes subunit vaccine against genus klebsiella using pan-genomics, reverse vaccinology, and biophysical techniques, *Vaccines* 9 (10) (2021) 1087, <http://dx.doi.org/10.3390/vaccines9101087>.
- [17] J. Ma, S. Wang, C. Zhao, X. Yan, Q. Ren, Z. Dong, J. Qiu, Y. Liu, Q. Shan, M. Xu, B. Yan, S. Liu, Computer-aided discovery of potent broad-spectrum vaccine adjuvants, *Angew. Chem. Int. Ed.* 62 (18) (2023) <http://dx.doi.org/10.1002/anie.202301059>.
- [18] R. Rappuoli, Reverse vaccinology, *Curr. Opin. Microbiol.* 3 (5) (2000) 445–450.
- [19] X. Yang, Y. Wang, R. Byrne, G. Schneider, S. Yang, Concepts of artificial intelligence for computer-assisted drug discovery, *Chem. Rev.* 119 (18) (2019) 10520–10594, <http://dx.doi.org/10.1021/acs.chemrev.8b00728>.
- [20] S. Thomas, A. Abraham, J. Baldwin, S. Piplani, N. Petrovsky, Artificial intelligence in vaccine and drug design, *Methods Mol. Biol. (Clifton, N.J.)* 2410 (2022) 131–146.
- [21] C.R.B. Bonin, G.C. Fernandes, R.W. Dos Santos, M. Lobosco, Mathematical modeling based on ordinary differential equations: A promising approach to vaccinology, *Hum. Vaccines Immunother.* 13 (2) (2017) 484–489.
- [22] C.R.B. Bonin, G.C. Fernandes, R. de Menezes Martins, L.A.B. Camacho, A. Teixeira-Carvalho, L.M.H. da Mota, S.M.B. de Lima, A.C. Campi-Azevedo, O.A. Martins-Filho, R.W. Dos Santos, et al., Validation of a yellow fever vaccine model using data from primary vaccination in children and adults, re-vaccination and dose-response in adults and studies with immunocompromised individuals, *BMC Bioinformatics* 21 (2020) 1–25.
- [23] A.S. Perelson, Modeling the interaction of the immune system with HIV, in: C. Castillo-Chavez (Ed.), *Mathematical and Statistical Approaches to AIDS Epidemiology*, in: *Lecture Notes in Biomathematics*, vol. 83, Springer Berlin Heidelberg, 1989, pp. 350–370.
- [24] C. Baker, G. Bocharov, C. Paul, Mathematical modelling of the interleukin-2 T-cell system: A comparative study of approaches based on ordinary and delay differential equation, *Comput. Math. Methods Med.* 1 (2) (1997) 117–128.
- [25] S.T. Chang, J.J. Linderman, D.E. Kirschner, Multiple mechanisms allow mycobacterium tuberculosis to continuously inhibit MHC class II-mediated antigen presentation by macrophages, *Proc. Natl. Acad. Sci. USA* 102 (12) (2005) 4530–4535.
- [26] Y. Vodovotz, C.C. Chow, J. Bartels, C. Lagoa, J.M. Prince, R.M. Levy, R. Kumar, J. Day, J. Rubin, G. Constantine, et al., In silico models of acute inflammation in animals, *Shock* 26 (3) (2006) 235–244.
- [27] C. Bonin, R.W. dos Santos, G. Fernandes, M. Lobosco, Computational modeling of the immune response to yellow fever, *J. Comput. Appl. Math.* 295 (2016) 127–138.
- [28] C.R.B. Bonin, G.C. Fernandes, R.W. dos Santos, M. Lobosco, A simplified mathematical-computational model of the immune response to the yellow fever vaccine, in: *Bioinformatics and Biomedicine (BIBM)*, 2017 IEEE International Conference on Bioinformatics and Biomedicine (BIBM), IEEE, 2017, pp. 1425–1432.
- [29] G. Pettet, H. Byrne, D. McElwain, J. Norbury, A model of wound-healing angiogenesis in soft tissue, *Math. Biosci.* 136 (1) (1996) 35–63.
- [30] B. Su, W. Zhou, K. Dorman, D. Jones, Mathematical modelling of immune response in tissues, *Comput. Math. Methods Med.* 10 (1) (2009) 9–38.
- [31] J.A. Flegg, H.M. Byrne, M.B. Flegg, D.S. McElwain, Wound healing angiogenesis: the clinical implications of a simple mathematical model, *J. Theoret. Biol.* 300 (2012) 309–316.
- [32] A.B. Pigozzo, G.C. Macedo, R.W. Dos Santos, M. Lobosco, On the computational modeling of the innate immune system, *BMC Bioinformatics* 14 (6) (2013) S7.
- [33] B.d.M. Quintela, R.W. dos Santos, M. Lobosco, On the coupling of two models of the human immune response to an antigen, *BioMed Res. Int.* 2014 (2014).
- [34] D.L. Chao, M.P. Davenport, S. Forrest, A.S. Perelson, A stochastic model of cytotoxic T cell responses, *J. Theoret. Biol.* 228 (2) (2004) 227–240.
- [35] M.P. Xavier, C.R. Bonin, R.W. dos Santos, M. Lobosco, On the use of gillespie stochastic simulation algorithm in a model of the human immune system response to the yellow fever vaccine, in: *2017 IEEE International Conference on Bioinformatics and Biomedicine, BIBM*, 00, 2017, pp. 1476–1482, <http://dx.doi.org/10.1109/BIBM.2017.8217880>, URL doi.ieeecomputersociety.org/10.1109/BIBM.2017.8217880.
- [36] F. Celada, P.E. Seiden, A computer model of cellular interactions in the immune system, *Immunol. Today* 13 (2) (1992) 56–62.
- [37] D. Morpurgo, R. Serenthà, P.E. Seiden, F. Celada, Modelling thymic functions in a cellular automaton, *Int. Immunol.* 7 (4) (1995) 505–516.

- [38] B. Kohler, R. Puzone, P.E. Seiden, F. Celada, A systematic approach to vaccine complexity using an automaton model of the cellular and humoral immune system: I. Viral characteristics and polarized responses, *Vaccine* 19 (7) (2000) 862–876.
- [39] M. Bernaschi, F. Castiglione, Design and implementation of an immune system simulator, *Comput. Biol. Med.* 31 (5) (2001) 303–331.
- [40] F. Pappalardo, G. Russo, M. Pennisi, G. Sgroi, G.A.P. Palumbo, S. Motta, D. Maimone, F. Chiacchio, Agent based modeling of relapsing multiple sclerosis: A possible approach to predict treatment outcome, in: 2018 IEEE International Conference on Bioinformatics and Biomedicine, BIBM, IEEE, 2018, pp. 1380–1385.
- [41] R. Laubenbacher, A. Niarakis, T. Helikar, G. An, B. Shapiro, R.S. Malik-Sheriff, T. Sego, A. Knapp, P. Macklin, J.A. Glazier, Building digital twins of the human immune system: toward a roadmap, *NPJ Digit. Med.* 5 (1) (2022) 64.
- [42] R. Laubenbacher, B. Mehrad, I. Shmulevich, N. Trayanova, Digital twins in medicine, *Nat. Comput. Sci.* 4 (3) (2024) 184–191, <http://dx.doi.org/10.1038/s43588-024-00607-6>.
- [43] K. Willcox, B. Segundo, The role of computational science in digital twins, *Nat. Comput. Sci.* 4 (3) (2024) 147–149, <http://dx.doi.org/10.1038/s43588-024-00609-4>.
- [44] H. Helgers, A. Hengelbrock, A. Schmidt, J. Strube, Digital twins for continuous mRNA production, *Processes* 9 (11) (2021) 1967.
- [45] Y. Uwamino, K. Nagashima, A. Yoshifuji, S. Suga, M. Nagao, T. Fujisawa, M. Ryuzaki, Y. Takemoto, H. Namkoong, M. Wakui, et al., Estimating immunity with mathematical models for SARS-CoV-2 after COVID-19 vaccination, *Npj Vaccines* 8 (1) (2023) 33.
- [46] S.J. Rhodes, G.M. Knight, D.E. Kirschner, R.G. White, T.G. Evans, Dose finding for new vaccines: The role for immunostimulation/immunodynamic modelling, *J. Theoret. Biol.* 465 (2019) 51–55.
- [47] S. Rhodes, N. Smith, T. Evans, R. White, Identifying COVID-19 optimal vaccine dose using mathematical immunostimulation/immunodynamic modelling, *Vaccine* 40 (49) (2022) 7032–7041.
- [48] M.P. Xavier, L.T. Pompei, A.C.G.d.O. Vieira, M.A. de Paula, C.R. Bonin, R.F. Reis, A.B. Pigozzo, B.d.M. Quintela, R.W. Dos Santos, M. Lobosco, Modelling the human immune system response to the ChAdOx1 nCoV-19 vaccine, in: 2021 IEEE International Conference on Bioinformatics and Biomedicine, BIBM, IEEE, 2021, pp. 3327–3333.
- [49] A. Flaxman, N.G. Marchevsky, D. Jenkin, J. Aboagye, P.K. Aley, et al., Reactogenicity and immunogenicity after a late second dose or a third dose of ChAdOx1 nCoV-19 in the UK: a substudy of two randomised controlled trials (COV001 and COV002), *Lancet* 398 (10304) (2021) 981–990, [http://dx.doi.org/10.1016/s0140-6736\(21\)01699-8](http://dx.doi.org/10.1016/s0140-6736(21)01699-8).
- [50] M. Voysey, A. Flaxman, J. Aboagye, P.K. Aley, S. Belij-Rammerstorfer, S. Bibi, M. Bittaye, F. Cappuccini, S. Charlton, E.A. Clutterbuck, et al., Persistence of the immune response after two doses of ChAdOx1 nCoV-19 (AZD1222): 1 year of follow-up of two randomized controlled trials, *Clin. Exp. Immunol.* 211 (3) (2023) 280–287.
- [51] X. Yang, G. Yang, J. Chu, GraphCL-DTA: A graph contrastive learning with molecular semantics for drug-target binding affinity prediction, *IEEE J. Biomed. Heal. Inform.* 28 (8) (2024) 4544–4552, <http://dx.doi.org/10.1109/jbhi.2024.3350666>.
- [52] C. Elkan, K. Noto, Learning classifiers from only positive and unlabeled data, in: Proceedings of the 14th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, in: KDD08, ACM, 2008, pp. 213–220, <http://dx.doi.org/10.1145/1401890.1401920>.
- [53] Y. Fan, C. Zhang, X. Hu, Z. Huang, J. Xue, L. Deng, SGCLDGA: unveiling drug-gene associations through simple graph contrastive learning, *Brief. Bioinform.* 25 (3) (2024) <http://dx.doi.org/10.1093/bib/bbae231>.
- [54] X. Yang, G. Yang, J. Chu, The neural metric factorization for computational drug repositioning, *IEEE/ACM Trans. Comput. Biol. Bioinform.* 20 (1) (2023) 731–741, <http://dx.doi.org/10.1109/tcbb.2022.3144429>.
- [55] S. Sadeghi, J. Lu, A. Ngom, A network-based drug repurposing method via non-negative matrix factorization, *Bioinformatics* 38 (5) (2021) 1369–1377, <http://dx.doi.org/10.1093/bioinformatics/btab826>.
- [56] X. Yang, F. Zhao, T. Ren, C. Chen, K.T. Byrne, A.V. Danilov, R.C. Sears, P.S. Nelson, L.M. Coussens, G.B. Mills, Z. Xia, OmicsTweezer: A distribution-independent cell deconvolution model for multi-omics data, *Cell Genom.* (2025) 100950, <http://dx.doi.org/10.1016/j.xgen.2025.100950>.
- [57] R. Storn, K. Price, Differential evolution—a simple and efficient heuristic for global optimization over continuous spaces, *J. Global Optim.* 11 (4) (1997) 341–359, <http://dx.doi.org/10.1023/A:1008202821328>.
- [58] E.F. Adolph, The metabolism and distribution of water in body and tissues, *Physiol. Rev.* 13 (3) (1933) 336–371.
- [59] M.N. Sawka, S.N. Cheuvront, R. Carter, Human water needs, *Nutr. Rev.* 63 (suppl_1) (2005) S30–S39.
- [60] P.E. Watson, I.D. Watson, R.D. Batt, Total body water volumes for adult males and females estimated from simple anthropometric measurements, *Am. J. Clin. Nutr.* 33 (1) (1980) 27–39.
- [61] A. Ramirez, C. Cox, Improving on the range rule of thumb, *Rose-Hulman Undergrad. Math. J.* 13 (2) (2012) 1.
- [62] M. Voysey, S.A.C. Clemens, S.A. Madhi, L.Y. Weckx, P.M. Folegatti, et al., Safety and efficacy of the ChAdOx1 nCoV-19 vaccine (AZD1222) against SARS-CoV-2: an interim analysis of four randomised controlled trials in Brazil, South Africa, and the UK, *Lancet* 397 (10269) (2021) 99–111, [http://dx.doi.org/10.1016/s0140-6736\(20\)32661-1](http://dx.doi.org/10.1016/s0140-6736(20)32661-1).
- [63] D. Adalja, K. Kalita, L. Čepová, P. Patel, N. Mashru, P. Jangir, Arpita, Advancing truss structure optimization—A multi-objective weighted average algorithm with enhanced convergence and diversity, *Results Eng.* 25 (2025) 104241, <http://dx.doi.org/10.1016/j.rineng.2025.104241>.
- [64] G. Srikanth, D. Nimma, R.V.S. Lalitha, P. Jangir, N.V.S. Kumari, Arpita, Food security based marine life ecosystem for polar region conditioning: Remote sensing analysis with machine learning model, *Remote. Sens. Earth Syst. Sci.* 8 (1) (2024) 65–73, <http://dx.doi.org/10.1007/s41976-024-00172-6>.
- [65] M. Khishe, P. Jangir, Arpita, S.P. Agrawal, S.B. Pandya, A. Parmar, L. Abualigah, Innovative diversity metrics in hierarchical population-based differential evolution for <scp>PEM</scp> fuel cell parameter optimization, *Eng. Rep.* 7 (1) (2024) <http://dx.doi.org/10.1002/eng.213065>.
- [66] M. Aljaidi, N. Mashru, P. Patel, D. Adalja, P. Jangir, Arpita, S.B. Pandya, M. Khishe, MORIME: A multi-objective RIME optimization framework for efficient truss design, *Results Eng.* 25 (2025) 103933, <http://dx.doi.org/10.1016/j.rineng.2025.103933>.
- [67] P. Jangir, Arpita, S.P. Agrawal, S.B. Pandya, A. Parmar, S. Kumar, G.G. Tejani, L. Abualigah, A cooperative strategy-based differential evolution algorithm for robust PEM fuel cell parameter estimation, *Ionics* 31 (1) (2024) 703–741, <http://dx.doi.org/10.1007/s11581-024-05963-x>.
- [68] Y. Ren, H.F. Isleem, W.J.K. Almoghay, A.K. Hamed, P. Jangir, Arpita, G.G. Tejani, A.E. Ezugwu, A.A. Soliman, Machine learning-based prediction of elliptical double steel columns under compression loading, *J. Big Data* 12 (1) (2025) <http://dx.doi.org/10.1186/s40537-025-01081-1>.
- [69] B. Cuppens, B. Bivort, G. Maertens, L. Meheus, G. Leroux-Roels, M. Van Ranst, N. Hens, Mathematical modeling of hepatitis b virus infection: dynamics, treatment, and persistence, *PLoS Comput. Biol.* 11 (10) (2015) e1004789, <http://dx.doi.org/10.1371/journal.pcbi.1004789>.
- [70] A. Gonçalves, J. Bertrand, R. Ke, E. Comets, X. de Lamballerie, D. Malvy, A. Pizzorno, O. Terrier, M. Rosa Calatrava, F. Mentré, R.A. Smith, A.S. Perelson, J. Guedj, Timing of antiviral treatment initiation is critical to reduce SARS-CoV-2 viral load, *Front. Immunol.* 11 (2020) 591802, <http://dx.doi.org/10.3389/fimmu.2020.591802>.
- [71] I. den Braber, T. Mugwagwa, N. Vrisekoop, L. Westera, D.C. Müller, R.J. de Boer, E.H. Schrijver, G. Spierenburg, K. Gaiser, E. Mul, S.A. Otto, A. Ruiter, M.T. Ackermans, F. Miedema, J.A. Borghans, K. Tesselar, J.L. Nieves, E. Derhovanessian, G. Pawelec, M.D. Hazenberg, Maintenance of peripheral naive t cells is sustained by thymus output in mice but not humans, *Proc. Natl. Acad. Sci. USA* 109 (12) (2012) 4828–4833, <http://dx.doi.org/10.1073/pnas.1113746109>.
- [72] D. van der Horst, S. Klein, R. Heijink, S.M. van Ham, J.A. Borghans, R.J. de Boer, K. Tesselar, F. Miedema, D. van Baarle, Naive b cells in humans: a rapidly renewing, heterogeneous population, *Blood* 119 (25) (2012) 5301–5309, <http://dx.doi.org/10.1182/blood-2011-11-390989>.
- [73] J. Herman, W. Usher, SALib: an open-source python library for sensitivity analysis, *J. Open Source Softw.* 2 (9) (2017) 97, <http://dx.doi.org/10.21105/joss.00097>.
- [74] I.M. Sobol, Global sensitivity indices for nonlinear mathematical models and their Monte Carlo estimates, *Math. Comput. Simulation* 55 (1–3) (2001) 271–280.
- [75] S. A. R. M. A. T. C. F. C. J. G. D. S. M. T. S. Global Sensitivity Analysis: The Primer, Wiley, Chichester (England), 2008.
- [76] J. Nossent, P. Elsen, W. Bauwens, Sobol'sensitivity analysis of a complex environmental model, *Environ. Model. Softw.* 26 (12) (2011) 1515–1525.
- [77] T. Homma, A. Saltelli, Importance measures in global sensitivity analysis of nonlinear models, *Reliab. Eng. Syst. Saf.* 52 (1) (1996) 1–17.
- [78] P.M. Folegatti, K.J. Ewer, P.K. Aley, B. Angus, S. Becker, S. Belij-Rammerstorfer, D. Bellamy, S. Bibi, M. Bittaye, E.A. Clutterbuck, et al., Safety and immunogenicity of the ChAdOx1 nCoV-19 vaccine against SARS-CoV-2: a preliminary report of a phase 1/2, single-blind, randomised controlled trial, *Lancet* 396 (10249) (2020) 467–478.
- [79] J. Feinberg, H.P. Langtangen, Chaospy: An open source tool for designing methods of uncertainty quantification, *J. Comput. Sci.* 11 (2015) 46–57, <http://dx.doi.org/10.1016/j.jocs.2015.08.008>.
- [80] B. Genser, P.J. Cooper, M. Yazdanbakhsh, M.L. Barreto, L.C. Rodrigues, A guide to modern statistical analysis of immunological data, *BMC Immunol.* 8 (2007) 1–15.
- [81] D.E. Williams, Statistics of antibody binding to the spike protein explain the dependence of COVID 19 infection risk on antibody concentration and affinity, *Sci. Rep.* 12 (1) (2022) 9379.
- [82] F. Pukelsheim, The three sigma rule, *Amer. Statist.* 48 (2) (1994) 88–91.
- [83] C.C. Heyde, E. Seneta, I.J. Bienaymé: Statistical Theory Anticipated, Vol. 3, Springer Science & Business Media, 2012.
- [84] J.G. Saw, M.C. Yang, T.C. Mo, Chebyshev inequality with estimated mean and variance, *Amer. Statist.* 38 (2) (1984) 130–132.
- [85] Odeint, Odeint's homepage, 2020, URL <http://docs.scipy.org>.

- [86] D. Shafer, Z. Zhang, *Introductory Statistics Version 1.0*, Flat World Knowledge, Incorporated, 2012, URL <https://books.google.com.br/books?id=YzVcvgAACAAJ>.
- [87] V. Valim, O.A. Martins-Filho, M.d.P.G. Gouvea, L.A.B. Camacho, D.A.M. Vilela, S.M.B. de Lima, A.S. Azevedo, L.F.P. Neto, C.M.A.S. Domingues, N.F. de Medeiros Junior, et al., Effectiveness, safety, and immunogenicity of half dose ChAdOx1 nCoV-19 COVID-19 vaccine: Viana project, *Front. Immunol.* 13 (2022) 966416.
- [88] S. Nanthapisal, T. Puthanakit, P. Jaru-Ampornpan, R. Nantanee, P. Sodsai, O. Himananto, J. Sophonphan, P. Suchartlikitwong, N. Hiransuthikul, P. Angkasekwinai, et al., A randomized clinical trial of a booster dose with low versus standard dose of AZD1222 in adult after 2 doses of inactivated vaccines, *Vaccine* 40 (18) (2022) 2551–2560.